UNIVERSITI TUNKU ABDUL RAHMAN

FACULTY OF INFORMATION & COMMUNICATION TECHNOLOGY (KAMPAR CAMPUS)

FINAL YEAR PROJECT I (FYP I)

INFORMATION BOOKLET

Prepared by FYP Committee

***Version 16 October 2025**
***Note: Please always refer to the most updated version**

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Important Notice to All Students

Plagiarism is a serious offence. Copy and paste for the report content is prohibited.

You must sign the report submission declaration to confirm that your FYP report has been done by your own efforts without any plagiarism.

Please report any error found in this document to FICT DD of R&D directly or through your supervisor

1 Introduction

Every UTAR student undertaking a degree program in FICT is required to complete a project under the supervision of a FICT (Kampar) academic staff or an external supervisor from the industry. In the case whereby an external supervisor is appointed, a FICT (Kampar) staff shall be appointed as a co-supervisor for the student. The project should provide student with the opportunity to bring together the academic knowledge and skills acquired from the range of modules already studied. In general, the whole project can be divided into three parts, namely Introduction to Inventive Problem Solving and Proposal Writing (IIPSPW), Project I and Project II, which are to be completed by Year 3.

Although students are required to take the modules starting from Year 2 semester 2 or 3, they are encouraged to explore the areas of interest, identify the project supervisors and define a project topic if possible, as early as the first semester of Year 2. The detailed planning of the project is described in the following sections.

The objectives and learning outcomes are listed as below.

The objectives of Project I:

[1] To introduce a general approach in starting a project, the need for proper documentation and reporting, and professional presentation of work undertaken;

[2] To equip students with the relevant research and technical skills that may be utilized for the formulation and development of a project;

[3] To introduce current Information & Communication Technology (ICT) related trends and development.

The learning outcomes of Project I:

After completing this unit, students will be able to:

CLO 1 – Identify a topical or problem area of interest for an ICT final year project (FYP) together with its project plan, scope and objectives.

CLO 2 – Relate formal documentations, such as proposal, literature search summary, work log and report, required for the FYP.

CLO 3 – Determine suitable methods and tools for problem analysis and project development.

CLO 4 – Demonstrate formal presentation skills for a proposal and technical work.

CLO 5 – Organize strengthened and convincing proposal.

Generally, Projects I will involve elements of preliminary investigation, project design, realization of design and evaluation.

1.1 Selection of Project Area and Project Proposal

In Year 2 Semester 2 or 3, students are expected to register for Introduction to Inventive Problem Solving and Proposal Writing (IIPSPW). Among the objectives of this course is to equip students with relevant skills in planning, writing and presenting project proposals. At the end of this course, students will have identified and confirmed their project titles with the respective potential (FYP) supervisors together with a completed preliminary proposal report. Students are required to register their project title with the approval of an academic staff as the supervisor (by week 7).

Students who have taken IIPSPW and wish to modify their project titles or change supervisors when enrolling in FYP I may do so subjected to the agreements of the IIPSPW supervisor, the new supervisor for supervising FYP I and II projects and the FYP committee. (Please contact the Deputy Dean of R&D who chairs the FYP committee)

Students who have completed their FYP I should not change their project completely; for example, from an Android application development to a Webpage development. However, students can change their project title; for instance, the original project is about developing a retail POS system, and need to add or reduce the objectives or scope of the project due to additional modules can enhance the system or to reduce the size of the project. The change of project title at this stage is mainly to better describe the project objectives.

A starting point to tackle the final year project is the identification and selection of an area of interest. A session will be held in IIPSPW class whereby relevant lecturers are invited to give a short talk to students. This will expose students to the areas available and the people involved. To know better an area of work, students can seek out potential project supervisors for further discussion. Also, students can look into the Internet for more information on the areas of their own interests. After agreeing with a supervisor on the area of study (and perhaps with a tentative title), students can then perform background reading for their final year project endeavour.

A list of suitable project topics or areas offered by the lecturers is available at the faculty website at https://fict.utar.edu.my/FYP/fyp_offer_titles.php. Students are also encouraged to suggest their own projects or projects in collaboration with firms in the industry (where appropriate). To do so, students must prepare a draft proposal and discuss with the relevant lecturer to ensure that the proposal is of a suitable level and standard.

A FULL proposal is to be submitted at the middle of Final Year Project (Project I) for evaluation by the supervisor, after which students would start working toward the completion of a prototype of their projects. Students need to meet their individual supervisors on regular basis, at least, once per fortnight.

By the end of the Project I, students will hand in their project prototype for evaluation by the supervisor.

1.2 General Classification of Final Year Project

The following is a guide and framework for setting up and running your project. There are two broad categories:

- Research-based projects;
- Development-based projects.

These categories are merely a guide to help students design their projects. While most projects will fit neatly into one of these two categories; some projects may have characteristics of both of these categories. It is important for students to recognise what project category that their project might fit into so as to enable them to address the relevant learning outcomes and requirements in which will help defining clear and concise project objectives.

1.3 Research-based Projects

Academic research projects are undertaken to investigate a research question. An academic research project must contain a research contribution from the student, for example, the development of a model or the design of an algorithm towards analysing/solving a problem. A research project might include data gathering; however, the gathering of data in itself does not constitute an acceptable level of research effort by the student. Rather some rigorous analysis of the data and/or the development of some deliverable based on the data are required.

The deliverable should have the potential for further research used by a third party, for example the supervisor, an external body or other stakeholders in the project. Furthermore, academic research projects must be designed so that it is clear of what factors affect the validity and generalization of the results.

In developing Academic Research projects, your proposal should state:

- The research question to be addressed;
- Any research initiative or project that your project is a part of;
- The research methods and tools to be used;
- How you will judge the validity and generality of your results;
- In what ways the project may contribute to related research activities.

1.4 Development-based Projects

These types of projects involve design and construction of a prototype for an application that can be in the form of hardware or software or a mixed hardware/software. The design and construction must be non-trivial. The development should follow an established hardware/software engineering method. In exceptional cases, students are permitted to do projects that involve analysis and design without a construction. The outcome of such project should provide sufficient information for implementation by a third party in the future. Alternatively, a formal theory may be built and its soundness and application are demonstrated. The following should be stated for Application Development projects:

- The purpose of the hardware/software;
- In what way the project is novel;
- What theory (if any) underpins the project;
- Applicable hardware/software engineering methods;
- What tools will be used, so far as decided;
- Methods envisaged for testing and evaluating the hardware/software;
- How the complexity of the work merits it being a final year project.

1.5 Combining Project Categories

For projects that do not fit neatly into one of the two project categories, the union of the respective lists of details must be clearly stated in the proposal.

2 Pitfalls and Problems

The final year project will be a demanding but exciting learning experience. However, it is not without problems, if not identified and addressed, could seriously affect the final result and ultimately reduce your grade. This section addresses some of these problems and provides tips on how to avoid them.

a. ***The "Overachiever" Problem***. A common problem is selecting a topic that is far too ambitious for the allotted time. Remember that you have only 12-13 weeks to finish the coding, debugging, and testing. Be careful not to select a topic that is unrealistically large. This can lead to frustration as well as errors caused by "cutting corners" and hurrying through the implementation. Discuss with your supervisor the scale of what you are planning. If he or she thinks it may be too large, consider implementing the project in stages, and complete the project stage by stage. When one stage is working, then move on to the next stage. If you cannot complete the project, you will still have a functioning system.

b. ***The "Do It Tomorrow" Problem***. Thirteen weeks sounds like a long time, but it goes by quickly. You need an implementation schedule that allocates reasonable amounts of work throughout the entire semester. Then you must stick to that schedule. Don't be tempted to postpone work on the project because week 13 seems so far off. All that happens is that during the final few weeks you rush madly to get something working, and software implemented in a rush rarely works correctly!
REMEMBER: **Four marks will be deducted for all late submission.**

c. ***The "Sleeping Member" Problem***. In the ideal world, all team members have equal ability, equal interest in the problem, and work equally hard. This may not happen in the real world. You may have one or more team members who do not carry their share of the workload, not because of a lack of ability, but rather lack of interest or motivation. This is a serious problem because, although part of your grade is based on each individual's effort, another part is based on successfully finishing the project. A non-contributing team member can slow down or prevent completion of the work. If you have a teammate who is not doing his or her share of the work, talk to them and stress the importance of everyone's role. If this does not solve the problem then talk to your supervisor. Do not let the failure of others prevent you from completing the work and receiving a good grade.

d. ***The "Poop Out At The End" Problem***. You have worked hard for 13 weeks to complete this project. You have spent many late nights and caught hundreds of bugs, and the program is working, so are you done? Absolutely not! The project grade is not based only on the programs you developed but also on your written reports and oral presentations. Remember that even though you may be 'burned out" from implementation, there is still work to be done. Do not produce a poorly written document or give a poorly organized presentation. That will negate much of your good work. Put in the time needed to prepare both a well written, high-quality final report and a well-organized, polished presentation. In fact, you should write your project document starting on the first day. The project document is the blueprint of your project, so you should write the document before implementing your project. A good job on these last steps will ensure that you receive a grade that fairly represents the work you have done.

3 Project Report Contents and Arrangement

Students should not copy large sections of books and/or reports. By copying, the change in writing style and the way information is presented can be easily detected. Students will be penalised for copying. Whenever values of short passages have been quoted, the full reference should be given. Students will be penalised for not referencing previous work.

3.1 Report Contents and Arrangement Guidelines for Project I

The essential components of the content of the final year project proposal should include the items listed below. They should also be arranged in the top-down order as listed. The proposal usually should not exceed 80 pages. Please follow/copy the report writing example in Appendix D. Report with incorrect format can get 5 marks (1 grade) reduction.

Item No.	Arrangement of the Proposal	Content
1	Title Page	Refer to Appendix A.
2	Acknowledgements	Maximum 1 page. Usually, you should first thank those who helped you academically or professionally, such as your supervisor, funders, and other academics. Then you can include personal thanks to friends, family members, or anyone else who supported you during the process. Refer to Appendix A.
3	Copyright Statement	*Refer to Appendix A.
4	Abstract	Maximum 1 page. Usually, it is limited to between 150 and 300 words. *Also state the area of study (Minimum 1, Maximum 2) and the keywords(Minimum 5 , Maximum 10). Refer to Appendix A. It should describe the format / outline of the proposal. Abstract is a formal summary of your completed work: ● Abstract, like summary, covers the main points of a piece of writing that includes the field of study, problem definition, methodology adopted, research process, conclusion and planning of the project work, etc. ● Unlike executive summaries written for non-specialist audiences, abstracts use the same level of technical language and expertise found in the article itself. Abstract typically serves the following goals: ● Help readers decide if they should read an entire article. ● Help readers to see your key findings and achievement of your project. ● Help readers understand your project by acting as a pre-reading outline of key points. Help readers to review technical work without becoming bogged down in details.
5	Table of Contents	● Refer to Appendix A. It should list all the chapters and their corresponding sections and subsections found in the report.
6	List of Tables	● Refer to Appendix A. It should list all the tables and their corresponding

		page numbers found in the report.
7	List of Figures	• Refer to Appendix A. It should list all the figures and their corresponding page numbers found in the report.
8	List of Symbols	• Refer to Appendix A. It should list all the symbols found in the report and their corresponding meanings.
9	List of Abbreviations	• Refer to Appendix A. It should list all the abbreviations found in the report and their corresponding meanings.
10	Sectioning	The number of pages should be between 20 to 50 pages (excluding the tables and figures). Students should familiarize themselves with report writing skills such as division of work and report sectioning. Each chapter should begin on a new page. Within a chapter, separate the information pertaining to the chapter into sections and subsections. Subsection is limited to 3 levels only. The following is a general guideline on the arrangement of chapters and what to be included as part of each chapter, i.e. different supervisors might change the requirements and format as they see fit. **Chapter 1: Introduction (5 to 10 pages) • Problem Statement and Motivation (1 to 2 paragraphs) o With respect to the FYP title, what are the motivation and problem to be solved. o It should be short and concise, emphasizes on overview of problems and the motivation of the whole project. At the very minimum, students should present a summary of the problem and the problem domain of the project. o You need to justify the existence of your project. Problem statement - state the existing <u>problem to be solved</u>. Motivation - why you want to solve it, why the project is needed? Writing up on problem statement and motivation: you would like to solve some problems. You would like to improve something. You would like to develop something that does not existed or carry out enhancement on an existing work. Example: You would like to develop a voice recognition software, because the software does not exist. Or, you improve on an existing voice recognition software because the software often misinterprets some words. o Common mistake: students normally confuse problem statement (or motivation) with technical difficulties. • Project Scope (1-2 paragraph) o Describe what you are going to deliver at the end of the project. (e.g. a piece of software, a piece of hardware, an improvement plan of a system, a development framework, a research survey, a model of a system, or simulation result, etc.). Give a general overview of your solution of the problem. o Example: This project develops a model on the social behavior of Internet with various simulation results on some scenarios. This project involves a new algorithm design to speed up the grid computing. o Example 2: A novel system design plus its prototype. • Project Objectives (1-4 paragraph) o Describe the purpose and aims of the project which give more detailed information than the project scope.

- For example: The project aims to improve at least 10% in processing performance over the current Sun Solaris grid computing engine with our new algorithm.
- The following questions are applicable:
  - What in general will this project try to achieve?
  - What will this project focus on?
  - What IS NOT covered by this project?
- Common mistakes:
  - Stating learning objectives instead of project objectives. For example, learning programming languages, tools etc.
  - Treating project timelines as project objectives.

● Impact, significance and contribution (1-2 paragraph)

- Describe how the project is going to benefit the readers or anybody. Particularly, whether it can potentially change/benefit the society or world to create long term and significant effect.
- Why are the problem and solution of your project interesting?
- Why is your project worth your readers' time to read it?
- Make your readers feel that your project is important or "desirable".
- This is where you need to "sell" or "promote" your project.
- For example: By having this educational software, the student will visualize better on how the processor works.
- For example: This survey has to be carried out because it will form the basis to anticipate and project the market trend ahead of time.

● Background information (> 3 paragraphs)

- A brief section giving background information may be necessary, especially if your work spans two or more traditional fields.
- Give a descriptive view on the field (or sub-field) of the project and historical development prior to the project.
- Give your readers who may not have any experience with some of the material needed to follow your project.
- It may be a good practice to give some definition of some key terms, or impart some key technical knowledge to the readers at this point.
- The ultimate question: What my readers, especially those who are not the same field as I do, need to know before they continue to read the rest of the document?

**Chapter 2: Literature Review (5 to 20 pages)

● Literature Review

- Highlight what is the current practice or prior arts towards the problem. It can be structured or non-structured (for unexplored areas)

		o If there are prior arts, students should refer or cite them and include the referenced art in the references section. ● Fact Finding (if the nature of the study needs this analysis) o Scientific method to do fact finding and analysis - reviewing existing manuals and procedures, preparing questionnaires, observations, research and conducting personal interviews. o Accomplished by techniques such as data element analysis; input-output analysis, including flow diagrams; recurring data analysis; and report use analysis. ● Data Collection (if the nature of the study needs this analysis) o Collect relevant data and documents to justify the problems and need for solutions ● Critical Remarks of previous works o Describe the strength and weakness of any previous work that are similar to your project o Compare them with your proposed solutions. **Chapter 3: Proposed Method/Approach (3 to 20 pages) ● Design Specifications o Methodologies and General Work Procedures A brief statement (1 to 2 paragraphs) of the methodology for the realization of the project. It could define the general approach to how the project and its output(s) will be realized. Show diagrams if they can help you to explain. o Tools to use. o User requirements (if there is any) o System Performance Definition: targeted improvements like accuracy, timing, etc. o Verification Plan: types of inputs to test, etc. ● System Design / Overview o All development projects should have this section which describes in some detail how the project is designed. o Give top-down system design diagrams (i.e. system flowchart, design block diagram, etc.) if applicable. Explain what is the functionality (and any other detail) of each block in diagrams. o For example, if your project is about writing an Android application for time management, then you should mention about the system flow diagram, how your program is written, such as the classes and methods used in the program, the database for keeping information, such as the SQL for creating the database, how to compile the program, and how to upload the program to a phone. o Another example is if you are building a moving robot, then you should mention how the physical robot is built, the parts or components required, mention about the design of your program that controls the robot, how to program the microcontroller, such as the steps to compile and upload the program to the microcontroller, how the sensors, servos, wires are connected, provide a schematic diagram that illustrates the connection of all the components, and how to operate the robot. [Please refer to your supervisor regarding your subtopics, above is just a guideline.]

		● Implementation Issues and Challenges (1-2 paragraphs) o Difficult issues and challenges in the implementation. o Novel aspects of this project (if any) ● Timeline (1 paragraph) o Estimated timeline for deliverables and milestones o Graphical - Gantt chart format o Planning for current semester and next semester. **Chapter 4: Preliminary Work (1 to 3 pages) o Report the preliminary work done and results obtained according to the proposed method in Chapter 3. o Comment and highlight the feasibility of the proposed method according to the preliminary work. **Chapter 5: Conclusion (1 to 3 pages) o Summarize the project including the problem, motivation, and proposed solutions. o Highlight any derived novel idea if there is any. o No new ideas or information in the conclusion, i.e. they should have been mentioned in other chapters.
11	Reference	Refer to Appendix B. It should list all the reference materials used for the project. (10 to 50 references)
12	Appendices	The appendices are supplementary materials that are too lengthy to be included or may not suite well in the main part of your document. The following is a guideline on the arrangement of appendices and what may be included as part of the appendices. - Specifications, data sheets and drawings of equipment or components used. - Data used for analysis. - Survey sheets. - Charts and data tables. - Lengthy mathematical derivations. - etc.
13	Poster	See description in section 3.2.

Remark:
**** Please ensure that all highlighted text in your report, particularly in the Copyright Statement form and Abstract, is properly edited to reflect your own information. After making the necessary changes, you must remove all yellow highlighting before submitting your report. Highlighted text is only provided as a guide and should not appear in the final version of your submission.***
***** The content of the report structure (Item No. 10. Sectioning) can be re-organized; the section title and subsection title can be renamed; NOT necessary exactly follow the sequence provided by the guideline.***

3.2 Poster Content and Arrangement Guidelines for Project I

The essential components of the content of the poster report should include the items listed as follow:

Item No.	Poster Presentation	Content
1	Size	A4 paper
2	Font	Use contrasting fonts for the title, text and figure legends. (Ensure the font size

		used are large enough)
3	Required Elements	- You may use photos, figures, and table - Determine a logical sequence for the material you will be presenting. - Organize that material into sections, e.g., - FYP1: Introduction, Methods (may use flowchart or block diagram), Discussion, and Conclusions. - Arrange the material into columns.
4	File Type	Softcopy, save the softcopy in any of the following format: JPEG / TIFF / BMP / EPS.
5	See poster examples at FICT FYP web site, https://fypfict.utar.edu.my/	

3.3 FYP1 Submission Form for Project I

Students should complete the FYP1 Submission Form (https://fypfict.utar.edu.my/documents/FYP1_Submission_Form.docx) (see Appendix D) and submit it separately along with the Project I proposal report.

The FYP1 Submission Form includes:

- Final Year Project Bi-Weekly (Long Trimester) / Weekly (Short Trimester) Report Form (at least six bi-weekly/weekly report forms)
- Plagiarism Check Result
- Supervisor's Comments on Originality Report Generated by #Turnitin for Submission of Final Year Project Report (for Undergraduate Programmes): FM-IAD-005 Form
- Checklist for FYP1 Report Submission Form

Remark:
 # When submitting your report to Turnitin, please ensure that you upload only the main report content and the reference list. To obtain an accurate similarity score and prevent unnecessary matches, you are required to REMOVE sections such as the declaration, acknowledgments, table of contents, list of figures or tables or symbols or abbreviations, headers, footers, and appendices that contain forms, code, poster, or reused materials. The purpose of this submission is solely for similarity checking; therefore, ONLY main chapters or sections should be INCLUDED. Please remember to keep your complete report file intact for final submission after the Turnitin check.

4 Project Report Format

4.1 Report Format for Project I

The report should be written using the third person and in the past tense. For example, do not use "I" and "you" in the report.

- Font
  – Times New Roman, 12 points, 1.5 line spacing applies to figure caption, table caption, chapter headings, subheadings, and citations.
  – Exceptions:
  Header, Footer, Footnote, Words in Figure/Table, font size should be within 10 to 11 points.
  – Colour: black.
  – Citing references in text: refer to the citation style "IEEE Style Referencing" in Appendix B.

- Language
  – British English

- Header
  – Align left: chapter number and title.

- Footer
  - Align right: page number.
  - The following is to be aligned left:
    BIS (Honours) Information Systems Engineering
    Faculty of Information and Communication Technology (Kampar Campus), UTAR.

- Page Number
  - Align right at the Footer.
  - Title, Abstract, Table of Contents and Listing – pages are numbered using small Roman numeric (i, ii, iii, etc). Note even though the Title Page is numbered i, the number is not to be printed on the page.
  - Chapters and Bibliography – pages are numbered 1, 2, 3, etc.
  - Appendices – pages are numbered A-1, A-2, etc for Appendix A, B-1, B-2, etc for Appendix B and etc.

- Margins
  - Left (1 inch, except the front cover 1.2 inches)
  - Right (1 inch)
  - Header/Footer (0.5/0.4 inch)
  - Top/Bottom (1 inch)

- Title Page (refer to Appendix A)
  - Do not include UTAR logo.
  - The font used is Times New Roman 12.
  - Note the format (font type, size, capitalization and the sentences arrangement) of the Title Page in Appendix A must be strictly adhere to. Change the word “REPORT” to “PROPOSAL”.

- Table of Contents (refer to Appendix A)

- Tables/Figures (refer to Appendix A)
  - Should include table (figure) caption immediately below the table (figure).
  - Number the tables and figures sequentially, with respect to the chapter or section of a chapter. To be consistent, use either one format, not both.
  - For example, Table 2-2 is the second table of chapter 2.
  - For example, Table 4-2-6 is the sixth table of section 2 of chapter 4

- Citation: The way a reference is cited in text.
  - Use IEEE standard citation (please refer to Appendix B).

- Bibliography: The section containing all the references
  - Use IEEE standard citation (please refer to Appendix B).

4.2 Other Points to Note on Writing Report

1. A thesis should be written according to the intended group of reader. It should be in a logic form with strong explanation to convince the reader on the conclusion of the thesis. It should be written in good language and easy to understand. Any technical language or daily language should be avoided. As far as possible, all statements must be supported by facts, numbers, and data.

2. The writer should be able to defend all statements by referring to a reliable research or research findings.

3. Symbols or nomenclature used should be defined. Standard symbols or acronym normally accepted in engineering field can be used. International System Unit (S.I) should be used. If you use other units, SI equivalent unit should be in bracket.

4. Equations and formulae should be typed and in Italic. Avoid ambiguous equation that has alternative interpretations; for example:

   (y/x) = ax + b   is preferred compared to: y/x = ax + b

5. Diagram can include graphs and figures. Diagrams should be easy to understand.

   i. Every diagram should have relevant title and should be numbered using the Arabic number at the bottom (if possible different for each chapter).

ii. Coordinate units (abscissa) should be written clearly in the graph.
iii. All the data points and lines should be clear - generally it should not be more than 2 or 3 curves in every diagram.
iv. Types of different data points must be shown in a legend.
v. Every diagram should be referred and elaborated in the text.
vi. The gridlines should be in appropriate intervals.

5 Viva: Oral Presentation and Project Demonstration

This exercise is intended to assess the students' ability to deliver a technical presentation as a result of their project investigation. The Oral Presentation is attended and assessed by the Supervisor and Moderator.

The presentation should describe the aim of the project, an outline of the presentation, the results obtained and the extent to which the goals of the project are met. The time allocated for the presentation session is 15 to 20 minutes and an additional 10 minutes for the 'Question and Answer' session.

The project demonstration session can be arranged to be the subsequent session to the oral presentation session for effective assessment. Otherwise, the demonstration may be arranged separately. The time allocated for the demonstration session is not more than 30 minutes.

Marking Scheme FYP1

Proposal: **50%**
- Project Title (3%)
- Literature Review/ Background Studies/ Fact Finding (7%)
- Problem Statement, Objectives and Project Formulation (7%)
- Research Methods and/or Development Tools (10%)
- Project Plan (6%)
- Conclusion and Discussion (7%)
- Report Structure and References (5%)
- Language and Clarity of Writing (5%)

Presentation: **40%**
- Presenter's General Preparation (3%)
- Slide/poster Design (7%)
- Presentation Content (7%)
- Presentation Question and Answer Session (3%)
- Preliminary Work (Result) (10%)
- Innovation and/or Contributions (10%)

General Effort: **10%**
- Attitude towards people/working relationship (1%)
- Punctuality in report submission (4%)
- Ethical (1%)
- Independent, initiative, responsible (2%)
- Constantly updating supervisor of his/her work progress (2%)

Total: **100%**

6 FYP Guidelines for Supervisor and Moderators

Guidelines for Supervisor

The Project Student conducts his/her work under the direction of the Project Supervisor. The Supervisor can be a qualified internal academic faculty staff or someone qualified external to the faculty. In the case of an external supervisor, an internal supervisor will be attached to the project to act as the moderator.

The Supervisor's role is to stimulate discussion and indicate the various avenues of approach and resources available. Although the Supervisor may serve as a guide and mentor for the project, it is emphasized that the ultimate responsibility for the project lies with the students.

Purchase of special components or equipment requires prior consent from the Supervisor, who acts as the 'budget controller' due to the limited funds available.

The Supervisor will evaluate the biweekly report, the project proposal, the full report, the oral presentation and product demonstration.
If the student could not manage to meet the supervisor in 4 consecutive weeks, the lecturer will have to contact the student to find out the current status of the student, and report the situation to the Final Year Project committee.

Guidelines for Moderators

Moderator is member of staff whose function is to ensure a uniform standard of assessment is applied to each project.

Moderation will take place at two stages:

- An assigned Moderator will evaluate the oral presentation and product demonstration. The moderation forms are available in the faculty's shared drive.

Appendices

Appendix A: Sample of Report Arrangement

ONLINE B2B AND B2C PURCHASING

By

Anthony Chan Ming Wai

A PROPOSAL

SUBMITTED TO

Universiti Tunku Abdul Rahman

in partial fulfillment of the requirements

for the degree of

BACHELOR OF INFORMATION TECHNOLOGY (HONOURS)

COMPUTER ENGINEERING

Faculty of Information and Communication Technology
(Kampar Campus)

MAY 2024

ACKNOWLEDGEMENTS

I would like to express my sincere thanks and appreciation to my supervisors, XXXX and XXXX who has given me this bright opportunity to engage in an IC design project. It is my first step to establish a career in IC design field. A million thanks to you.

To a very special person in my life, Stephanie Yuen, for her patience, unconditional support, and love, and for standing by my side during hard times. Finally, I must say thanks to my parents and my family for their love, support, and continuous encouragement throughout the course.

COPYRIGHT STATEMENT

ABSTRACT

This project is regarding a growing trend – Internet of Things (IoT). IoT is a collection of multiple things that connected to the Internet and performing data collection, sharing, and processing. As the IoT trend is growing at an incredible speed, people begin to concern about the challenges raised from IoT. For example, security, user-friendliness, privacy, and customizable issues. There are two major problems will be the focus on this project – customisation and user-friendliness. As most of the existing IoT applications are unable to fulfil a user-friendly and customisable IoT monitoring system, there is still a huge learning gap for IoT users in terms of understanding and customising their IoT solution. Some researches and literature reviews are carried out to look deeper into IoT customisation and user-friendliness issues. The reviews of the products including IoT in a Box, EzBlock Studio, RaspController, IoTool, Lego Mindstorm EV3, SimpleLink SensorTag, Xiaomi Home and Skydrop Smart Sprinkler are written in the literature review section of the report. After reviewing the strengths and weaknesses of existing products, a clearer vision regarding the IoT challenges will be presented in the report. The solution of this project is to develop a customisable cloud-based IoT monitoring system with smartphone interface in order to assist users in learning IoT and customising preferable IoT solutions easily. The final product of the project is the combination of IoT hardware, Google Cloud Platform, Firebase and an Android mobile application with the ability to connect, control, monitor and customise users' IoT solutions.

Area of Study (Minimum 1, Maximum 2): Internet of Things, Computer Vision

Keywords (Minimum 5, Maximum 10): Data Collection in IoT, Monitoring Applications, Mobile Application

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LIST OF SYMBOLS

β	beta
Ω	Ohm (resistance)

LIST OF ABBREVIATIONS

CMOS	Complementary Metal Oxide Semiconductor
MOSFET	Metal Oxide Semiconductor Field Effect Transistor
IC	Integrated Circuit
DRC	Design Rule Checker
SCNA	Scalable CMOS N-Well Analog
ASIC	Application Specific Integrated Circuit
HDL	Hardware Description Language

Chapter 1

Introduction

In this chapter, we present the background and motivation of our research, our contributions to the field, and the outline of the thesis. ……….

1.1 Problem Statement and Motivation

Nowadays, criminal and victim identification, including two processes, identification (oneto-many matching), and verification (one-to-one matching), is an integral part of forensic investigation procedures. The identification process means the existing records in a database are compared with the biometric traits found at a crime scene. But the verification process means the biometric traits found at a crime scene are compared with those taken from an arrested suspect. ……

1.2 Research Objectives

The aim of the thesis is to propose new soft biometric traits and efficient algorithms for criminal and victim identification. Although their faces are masked, other body parts can be exposed. In this thesis, the leg geometry is proposed as one of soft biometric traits. ….

1.3 Project Scope and Direction

…………

1.4 Contributions

Our experiment and analysis confirm the feasibility of the proposed biometric features and methods for criminal and victim identification. Firstly, lower leg geometry is shown to be an effective soft biometric trait and it provides a foundation for further research on criminal and victim identification based on body geometry. Secondly, the alignment problem of androgenic hair patterns is handled by using hair follicles and leg geometry for high and low resolution images. Thirdly, the ……….

1.5 Report Organization

The details of this research are shown in the following chapters. In Chapter 2, some related backgrounds are reviewed. Then, a preliminary study of lower leg geometry as a soft biometric trait is presented in Chapter 3. And then, Chapter 4 describes an alignment algorithm with androgenic hair patterns to improve high and low resolution leg identification. Furthermore, Chapter 5 reports the ……

Appendix B: IEEE Style Referencing

Refer to UTAR library IEEE style referencing source
https://opac.utar.edu.my/gw_2013_2_8/html/default/en/Guides/IEEE-Reference-Guide-Online-v.04-20-2021.pdf

Or see the pdf file "IEEE Referencing Style" at FICT FYP Portal

Appendix C: Sample of Appendix

APPENDIX

Questionnaire Sample

1. Do you know what is CT scan and MRI scan image?
A. Yes
B. No
C. Heard of it, but not really understand it.

2. Do you know what is 3D reconstruction of a model?
A. Yes
B. No
C. Heard of it, but not really understand it.

3. Do you experiences in using any (AR) application (e.g Pokemon GO)?
A. Yes
B. No
C. Maybe

4. Do you know what is augmented reality technology?
A. Yes
B. No
C. Heard of it, but not really understand it.

5. Do you feel difficulties in understanding and visualizing the 2D photo or images of knee and ankle foot structure on the paper materials.
A. Yes
B. No

Code Sample

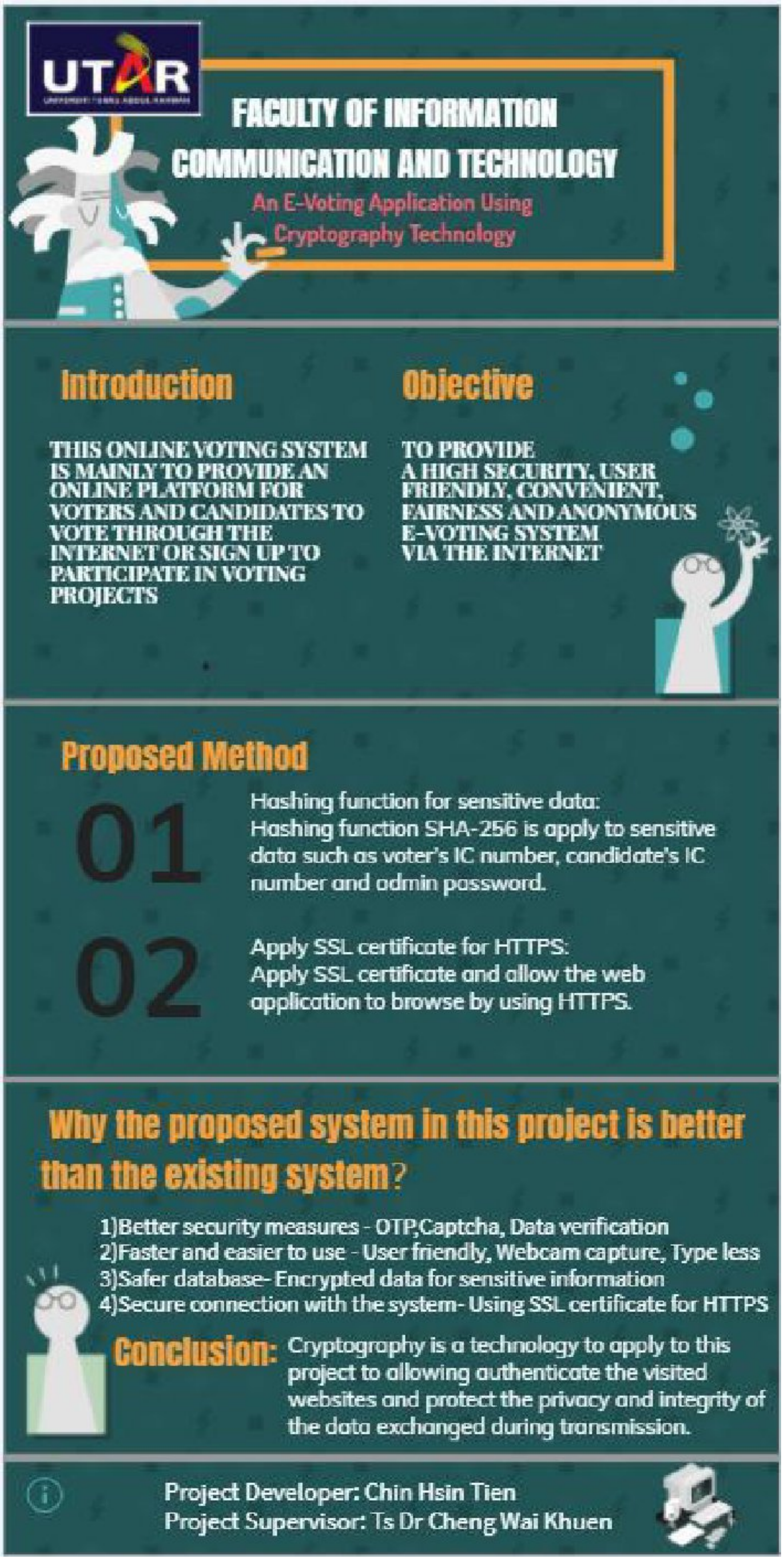
```
public class SUV extends Car {

	public SUV(double fuelCon, double engineLife) {
		super(fuelCon, engineLife);
		// TODO Auto-generated constructor stub
	}

	@Override
	public void useAddative(Addative ad) {
		// TODO Auto-generated method stub
		ad.effect(this);
		super.setEngineLife(super.getEngineLife() - 0.08);
	}

}
```

Poster


UTAR
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COMMUNICATION AND TECHNOLOGY
An E-Voting Application Using
Cryptography Technology
Introduction
THIS ONLINE VOTING SYSTEM IS MAINLY TO PROVIDE AN ONLINE PLATFORM FOR VOTERS AND CANDIDATES TO VOTE THROUGH THE INTERNET OR SIGN UP TO PARTICIPATE IN VOTING PROJECTS
Objective
TO PROVIDE A HIGH SECURITY, USER FRIENDLY, CONVENIENT, FAIRNESS AND ANONYMOUS E-VOTING SYSTEM VIA THE INTERNET
Proposed Method
01
Hashing function for sensitive data:
Hashing function SHA-256 is apply to sensitive data such as voter's IC number, candidate's IC number and admin password.
02
Apply SSL certificate for HTTPS:
Apply SSL certificate and allow the web application to browse by using HTTPS.
Why the proposed system in this project is better than the existing system?
1)Better security measures - OTP,Captcha, Data verification
2)Faster and easier to use - User friendly, Webcam capture, Type less
3)Safer database- Encrypted data for sensitive information
4)Secure connection with the system- Using SSL certificate for HTTPS
Conclusion:
Cryptography is a technology to apply to this project to allowing authenticate the visited websites and protect the privacy and integrity of the data exchanged during transmission.
Project Developer: Chin Hsin Tien
Project Supervisor: Ts Dr Cheng Wai Khuen

Appendix D: FYP1 Submission Form

FINAL YEAR PROJECT BI-WEEKLY (LONG TRIMESTER) / WEEKLY (SHORT TRIMESTER) REPORT

(Project I)

Trimester, Year:	**Study week no.:**
Student Name & ID:	
Supervisor:	
Project Title:	

1. WORK DONE [Please write the details of the work done in the last fortnight.]
2. WORK TO BE DONE
3. PROBLEMS ENCOUNTERED
4. SELF EVALUATION OF THE PROGRESS

Student's signature

Supervisor's signature

PLAGIARISM CHECK RESULT

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14% Overall Similarity

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- Quoted Text
- Small Matches (less than 8 words)

Exclusions

- 4 Excluded Sources
- 1 Excluded Match

Match Groups

37 Not Cited or Quoted 7%
Matches with neither in-text citation nor quotation marks

30 Missing Quotations 6%
Matches that are still very similar to source material

0 Missing Citation 0%
Matches that have quotation marks, but no in-text citation

0 Cited and Quoted 0%
Matches with in-text citation present, but no quotation marks

Top Sources

10% Internet sources
3% Publications
10% Submitted works (Student Papers)

Integrity Flags

0 Integrity Flags for Review

Our system's algorithms look deeply at a document for any inconsistencies that would set it apart from a normal submission. If we notice something strange, we flag it for you to review.

A Flag is not necessarily an indicator of a problem. However, we'd recommend you focus your attention there for further review.

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Match Groups

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- **30 Missing Quotations 6%**
  Matches that are still very similar to source material
- **0 Missing Citation 0%**
  Matches that have quotation marks, but no in-text citation
- **0 Cited and Quoted 0%**
  Matches with in-text citation present, but no quotation marks

Top Sources

10% Internet sources
3% Publications
10% Submitted works (Student Papers)

Top Sources

The sources with the highest number of matches within the submission. Overlapping sources will not be displayed.

1	Internet	www.geeksforgeeks.org	3%
2	Internet	eprints.utar.edu.my	2%
3	Internet	lds.csom.umn.edu	<1%
4	Student papers	City University of Hong Kong on 2025-07-04	<1%
5	Student papers	Universiti Tunku Abdul Rahman on 2023-04-23	<1%
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9	Internet	jmlr.org	<1%
10	Student papers	Somaiya Vidyavihar on 2024-10-27	<1%

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Form Title : Supervisor's Comments on Originality Report Generated by Turnitin for Submission of Final Year Project Report (for Undergraduate Programmes)			
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FACULTY OF INFORMATION AND COMMUNICATION TECHNOLOGY

Full Name(s) of Candidate(s)	
ID Number(s)	
Programme / Course	
Title of Final Year Project	

Similarity	**Supervisor's Comments** **(Compulsory if parameters of originality exceed the limits approved by UTAR)**
Overall similarity index:_________ % **Similarity by source** Internet Sources: __________ % Publications: __________ % Student Papers: __________ %	
Number of individual sources listed of more than 3% similarity: _________	
Parameters of originality required and limits approved by UTAR are as Follows: **(i) Overall similarity index is 20% and below, and** **(ii) Matching of individual sources listed must be less than 3% each, and** **(iii) Matching texts in continuous block must not exceed 8 words** ***Note: Parameters (i) – (ii) shall exclude quotes, bibliography and text matches which are less than 8 words.***	

Note Supervisor/Candidate(s) is/are required to provide softcopy of full set of the originality report to Faculty/Institute

Based on the above results, I hereby declare that I am satisfied with the originality of the Final Year Project Report submitted by my student(s) as named above.

____
Signature of Supervisor

Name: __________________________

Date: ___________________________

____
Signature of Co-Supervisor

Name: __________________________

Date: ___________________________

UNIVERSITI TUNKU ABDUL RAHMAN

FACULTY OF INFORMATION & COMMUNICATION TECHNOLOGY
(KAMPAR CAMPUS)

CHECKLIST FOR FYP1 REPORT SUBMISSION

Student ID	
Student Name	
Supervisor Name	

TICK (√)	**DOCUMENT ITEMS** Your report must include all the items below. Put a tick on the left column after you have checked your report with respect to the corresponding item.
	Title Page
	Copyright Statement
	Acknowledgement
	Abstract (with Area of Study and Keywords)
	Table of Contents
	List of Figures (if applicable)
	List of Tables (if applicable)
	List of Symbols (if applicable)
	List of Abbreviations (if applicable)
	Chapters / Content
	References
	All references are cited in the report, especially in the chapter of literature review
	Appendices (if applicable)
	Poster
	I agree 5 marks will be deducted due to incorrect format, declare wrongly the ticked of these items, and/or any dispute happening for these items in this report.

I, the author, have checked and confirmed all the items listed in the above table are included in my report.

____
(Signature of Student)

Date:

- End -